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Product inquiries 1644-4007

INDUSTRIAL CLEANING SOLUTION

BYCLEAN SERIES

BCS



BYCHEM

Dedicated to producing high-quality eco-friendly paints and cleaning agents, BYCHEM Co., Ltd. strives to provide quality and service to satisfy our customers.



Cleaning Agent Business Division

Low-toxicity, eco-friendly alternative industrial cleaning agent based on non-toxic agents



Refinement Business Division

Waste organic solvent refinement using high-purity treatment unit



Paint Business Division

Construction paint and Low-VOC, Low-toxicity diluent





Company History

2016 ~ Present — Grow into a global leader

- 2023** • Labor-Management-Civil-Government Joint Declaration Implementation Agreement
Model Taxpayer Certification
- 2022** • Certified as a faithful taxpayer
Commissioner of Employment and Labor Award
Gyeonggi Service Center completed
Promising small and medium business certification
ESG level 3
- 2021** • Ministry of Employment and Labor Youth-Friendly Small Giant Enterprise
A SME fostering talents
Patent for heat treatment method of aluminum alloy
Selected as Chungnam Star Enterprise
Selected as a promising export SME
Excellent businessman award
A famous root company
Minister of Trade, Industry and Energy Award for Corporate Innovation
Minister of SMEs and Startups Award commendation
- 2020** • Venture business confirmation
Excellent technology company
Good Ppuri Company to work at
Model Taxpayer Certification
- 2019** • Ministry of Employment and Labor Youth-Friendly Small Giant Enterprise
Environmental label certification
National Defense Venture Business
Good Company to Work for
Excellent work innovation company
Material/parts specialist company
Family-friendly business
Ppuri Company
Venture Business Administration Director Award
- 2018** • Company with excellent corporate capabilities
Technology Innovation Small and Medium Business (Innobiz)
- 2017** • Ministry of Employment and Labor Small but strong companies
Patent for non-chlorine cleaning agent composition
Environmental label certification
- 2016** • Completed Yesan Factory

1977 ~ 2015 — Shared future together

- 2015** • Sincere Taxpayer
- 2014** • Best Ansan Small and Medium Business / Industrial Accident Prevention Sector
Ministry of Employment and Labor Youth-Friendly Small Giant Enterprise
- 2012** • Metal cleaning and manufacturing method patent
- 2004** • Patent for hydraulic coating composition
- 2003** • China Chemical Quality MA Certification
- 2002** • Emulsion thermal insulation paint composition patent
China Environmental Green Building Materials Product Certificate
ISO 9001, 14001
Established China Corporation in Shanghai
- 2001** • NT Mark (new technology)
- 2000** • Changed company name (Bychem Co., Ltd.)
- 1999** • KS mark permission certification
- 1998** • Promising Small and Medium Business in Gyeonggi Province
- 1996** • ISO9002 certification
- 1992** • Technology R & D Center
- 1987** • Designated as a quality assurance company
- 1986** • Conversion to corporation
- 1977** • Founding of Baekyang Paint Chemical Industry Co.,Ltd



—★★★—
Patented

Incombustible alternative cleaning agent
[Determined as non-dangerous goods]

BCS-5000



Replacing TCE / 1,2-DCP toxic agents
Replacing NPBr reproduction-toxicity
/ Special control substance

Occupational Safety
and Health Act

Chemical Substances
Control Act

Safety Control of
Hazardous Substances Act

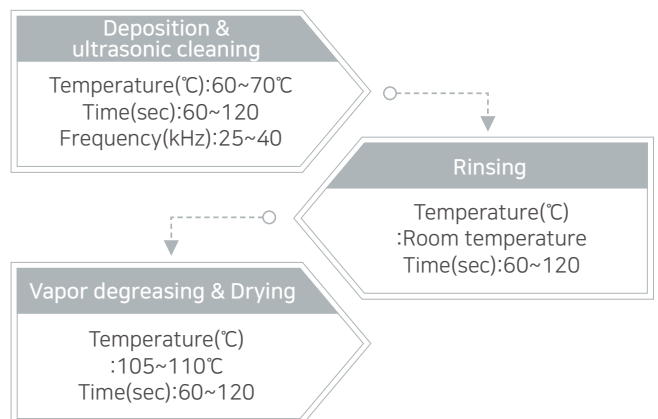
Not Applicable



Purpose of product		Applicable material		
Cleaning metal-working fluid		Automobile parts		
Cleaning PCB flux		Electric/electronic parts		
Cleaning grease and inhibited oil		Mobile phone parts		
Cleaning drawing oil and rolling oil		Metal-processing parts		
Other oil (preliminary test)		Other alloys (preliminary test)		

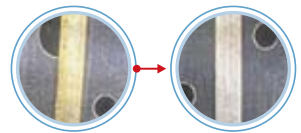
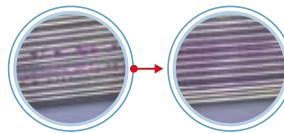
Category	BCS-5000	TCE	NPBr	1,2-DCP
Specific gravity	1.485	1.464	1.354	1.156
Distillation range(°C)	92~110	85~88	70~72	90~95
Flash point(°C)	>250	>250	69	16
Moisture content(%)	0.058	<0.1	<0.1	<0.1
Evaporation residue (%)	0.0013	<0.01	<0.01	<0.01

Product specifications : Drum (250kg), Pail 20L (23kg), CAN 18L (23kg)

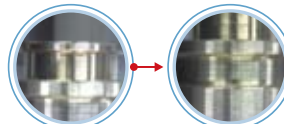


3~5-bath Ultrasonic & Vapor degreasing cleaning system

[Drawing oil / SUS Tube] [Punching oil / Nickel Silver]



[Cutting oil / Brass]



[Abrasive / Copper]



Cleaning agent to remove high-viscosity
drawing oil and rolling oil

BCS-5000(HP)



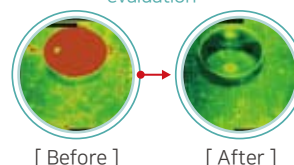
- ✓ Replacing TCE, MC, 1, 2-DCP, chloroform, and NPBr
- ✓ **Non-flammable at room temperature – Flash point above 100°C, patent applied**
- ✓ Excellent cleaning power for high-viscosity oils and waxes
- ✓ Established recycling purification system (patent applied) – Recyclable

BCS-5000(HP) Washability Test

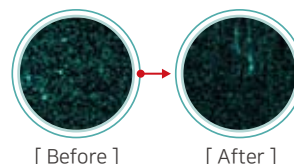
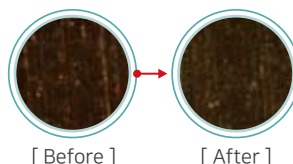
Cleanability evaluation using surface oil detector –
Recognoil 3W

BCS-5000(HP) / Aluminum
material / Punching oil cleaning
evaluation

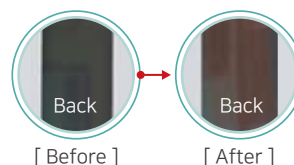
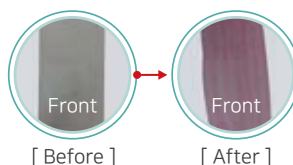
BCS-5000(HP) / Optical lens
/ Processing oil cleaning
evaluation



Cleaning evaluation by SEM-EDS analysis –
Applied to copper

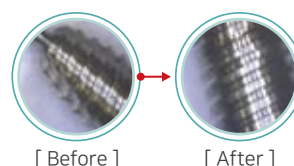


Cleaning evaluation by Dyne Pen Test –
Applied to the battery cell



SMT flux cleaning performance
evaluation – **3-tank ultrasonic
cleaning process applied**

Implant material cleaning
evaluation – **Metal cleaner
composition**





Low boiling point alternative cleaning agent
[Eco-friendly certified]

BCS-3000



Standard: EL605.Industrial cleaner

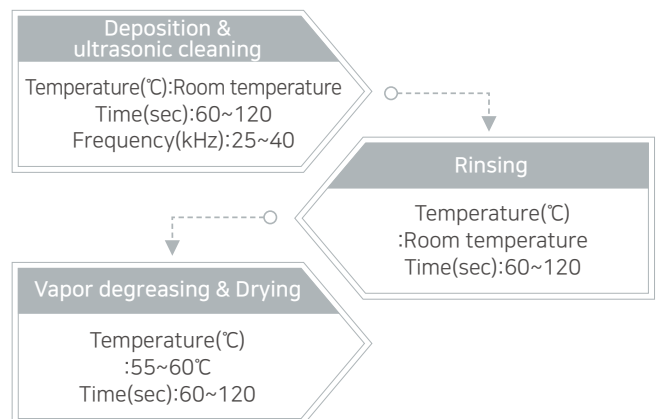
MC(Dichloromethane)
Toxic Substitution Replaces HCFC-141b and ozone-depleting substances



Purpose of product	Applicable material
Cleaning metal-working fluid	Automobile parts
Cleaning PCB flux	Electric/electronic parts
Wax and oil cleaning	Optical material components
High viscosity processing oil cleaning	Metal-processing parts
Other oil (preliminary test)	Other alloys (preliminary test)

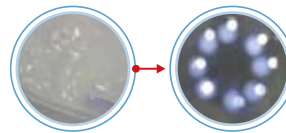
Category	BCS-3000	MC	HCFC-141B
Specific gravity	1.250~1.296	1.326	1.250
Distillation range(°C)	48~55	40~42	30~34
Flash point(°C)	>250	>250	69
Moisture content(%)	0.058	<0.1	<0.1
Evaporation residue (%)	<0.01	<0.01	<0.01

Product specification : Drum (200kg), Pail 20L (20kg), CAN 18L (20kg)

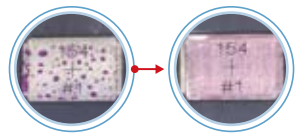


3~5-bath Ultrasonic & Vapor degreasing cleaning system

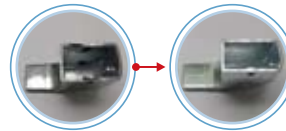
[WAX / Lenz]



[Punching oil / Nickel Silver]



[Cutting oil / SUS 304]

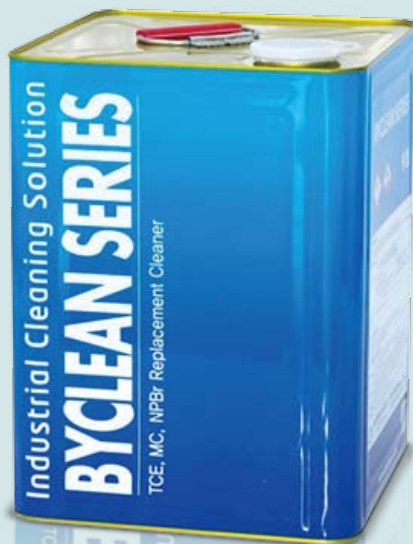


[Abrasive / Brass]



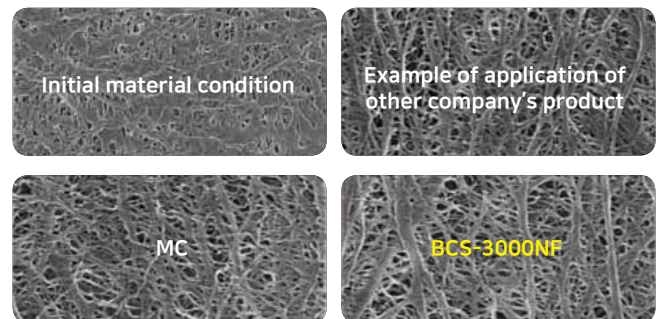
Secondary battery separator cleaner & electronic parts cleaner

BCS-3000NF



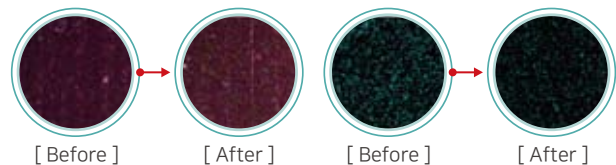
- ✓ Replacing methylene chloride (MC) toxic substances
- ✓ **Non-flammable and non-explosive – Certified by INERIS explosion test**
- ✓ Not subject to per- and polyfluoroalkyl substance (PFAS) regulations
- ✓ High cleaning performance, fast evaporation, and recycling system

▪ Example of application of removal of pore forming agent from battery separator material

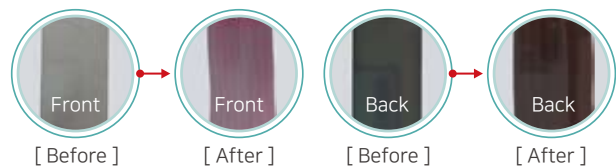


BCS-3000NF Washability Test

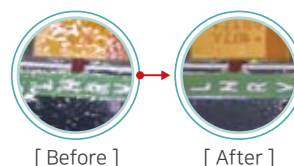
Cleaning evaluation by SEM-EDS analysis
– Applied to SUS-304



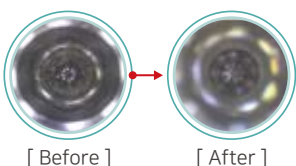
Cleaning evaluation by SEM-EDS analysis
– Applied to PP material



SMT flux cleaning performance evaluation – 3-tank ultrasonic cleaning process applied



Implant material cleaning evaluation – Metal cleaner composition



Special Purpose Alternative Cleaning Agent
[Flame Retardant]

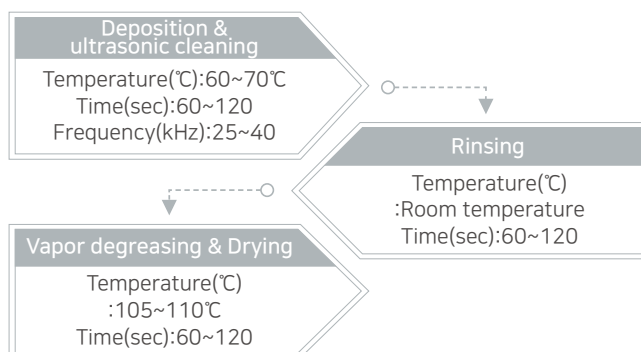
BCS-NEW1000 & Special products



- BCS-NEW1000 series : Eco-friendly, economic
- BCS-CC : Degreasing agent for drawing oil and rolling oil room temperature cleaning
- BCS-SS : IPA (isopropyl alcohol) Alternative low toxicity cleaner
- BCS-UT3 : Polyetherimide(ULTEM) exclusive cleaning agent

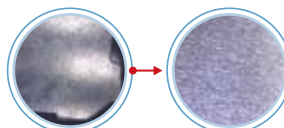
Category	BCS NEW1000	BCS NEW1000 (N)	BCS NEW1000 (D)
Specific gravity	1.331	1.247	1.264
Distillation range(°C)	90~110	90~110	90~110
Occupational Safety and Health Act	Uncomfortable	Controlled or regulated	Controlled or regulated
Chemical Substances Control Act	Not applicable(N/4)	Not applicable(N/4)	Not applicable(N/4)
Packing(CAN18L)	20kg	20kg	20kg
Packing(PE 20L)	23kg	-	23kg
Packing(drum)	230kg	220kg	230kg

Category	BCS - UT3	BCS - CC	BCS - SS(Alcohol)
Specific gravity	1.080	1.181	0.816
Distillation range(°C)	90~110	110	42~78
Occupational Safety and Health Act	Not applicable(N/4)	Controlled or regulated	Not applicable(N/4)
Chemical Substances Control Act	Not applicable(N/4)	Not applicable(N/4)	Not applicable(N/4)
Packing(CAN18L)	-	20kg	-
Packing(PE 20L)	20kg	-	16kg
Packing(drum)	200kg	200kg	160kg

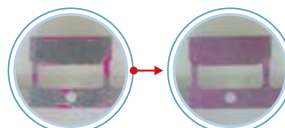


3~5-bath Ultrasonic & Vapor degreasing cleaning system

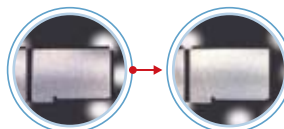
[Punching oil / Nickel Silver]
BCS-NEW1000



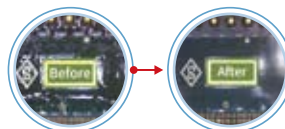
[Cutting oil / Aluminum]
BCS-NEW1000(N)



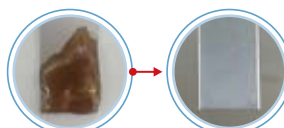
[Punching oil / Nickel Silver]
BCS-NEW1000(D)



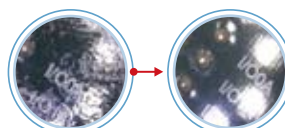
[Flux / PCB]
BCS-UT3



[Drawing oil / SUS 304]
BCS-CC

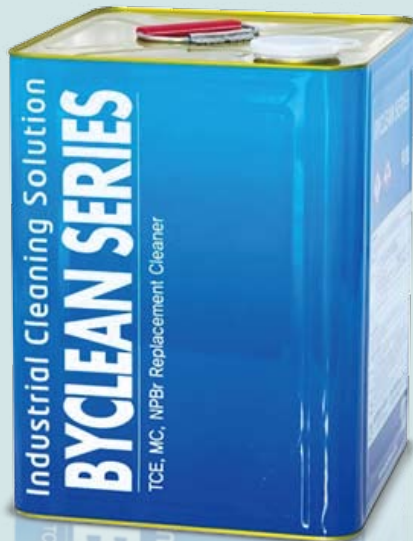


[Flux / PCB]
BCS-SS



Cleaning agent to remove high-viscosity
drawing oil and rolling oil

BCS NEW1000(HP)



- ✓ Replacing TCE, MC, 1, 2-DCP, chloroform and NPBr
- ✓ **Non-flammable at room temperature – Flash point above 100°C, patent applied**
- ✓ Excellent cleaning power for high-viscosity oils and waxes
- ✓ Established recycling purification system (patent applied) – Recyclable

Comparison before and after applying BCS-NEW1000(HP) cleaning agent(1)

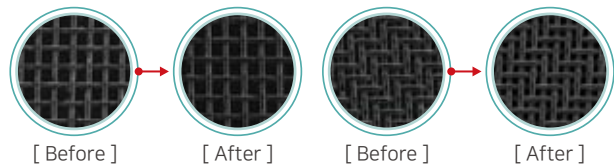


Comparison before and after applying BCS-NEW1000(HP) cleaning agent(2)

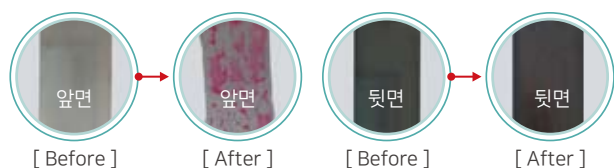


BCS-NEW1000(HP) Washability Test

Cleaning evaluation by SEM-EDS analysis
– Applied to PP material

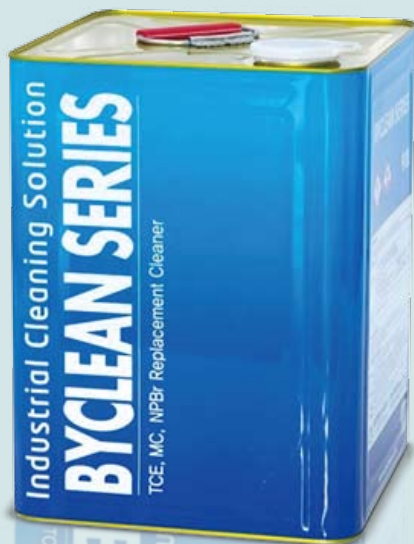


Cleaning evaluation by Dyne Pen Test
– Applied to the battery cell



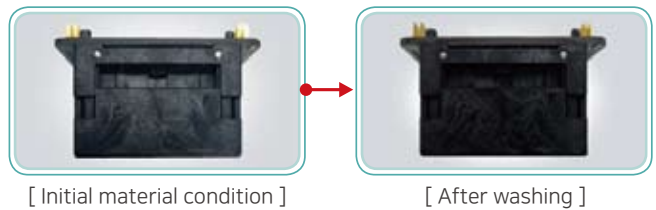
Cleaning agent for polyetherimide materials and semiconductor sockets

BCS-UT(HP)

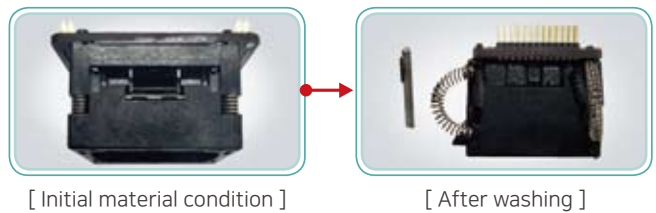


- ✓ Replacing heptane and hydrocarbon-based flammable cleaning agents
- ✓ **Non-flammable at room temperature – Flash point above 100°C, patent applied**
- ✓ Excellent cleaning performance for flux and solder paste
- ✓ Capable composition of stable chemical resistance to Ultem materials

▪ Comparison before and after application of BCS-UT(HP) cleaning agent

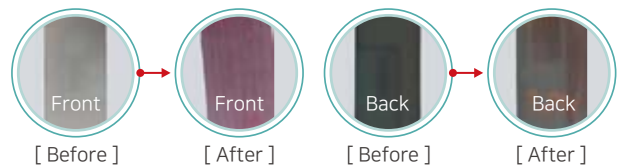


▪ Comparison before and after application of other company's cleaning agent

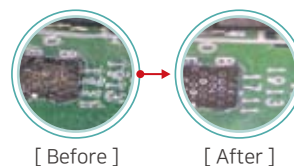


BCS-UT(HP) Washability Test

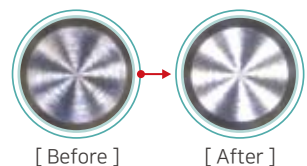
Cleaning evaluation by Dyne Pen Test
– Applied to the battery cell



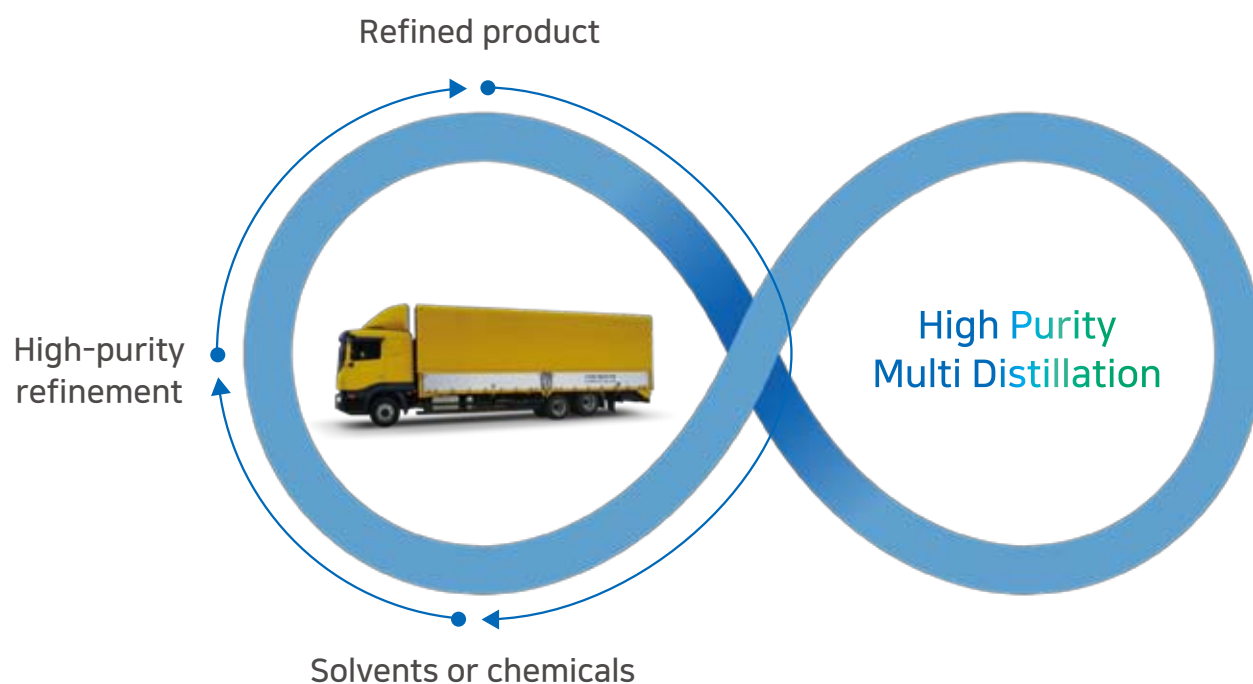
SMT flux cleaning performance evaluation – 3-tank ultrasonic cleaning process applied



Implant material cleaning evaluation – Metal cleaner composition



High-purity refinement recycling by fractional distillation system



Zero toxic agent & Zero waste

BYCHEM challenges to create better future for people and environment



Technical products
focused on safety &
environment



Zero toxic agent flame
retardant/incombustible
cleaning agent



Production-automated
smart factory



Expert consulting on
laws related to
chemical products